

Biomaterials

生物材料

南方科技大学材料科学与工程系2024年春季课程

Bio-inertness vs. Bioactivity

Bioactive materials **play a more aggressive role** in the body. While a biocompatible material should affect the equilibrium of the body as little as possible, a bioactive material recruits specific interactions between the material and surrounding tissue.

Bioinert materials refer to any materials that once placed in the human body **has minimal interactions** with its surrounding tissue, examples of these are stainless steel, titanium, alumina, partially stabilized zirconia, and ultra high molecular weight polyethylene.

Bioactive Materials

- Encourage tissue integration to aid in the fixation of an implant in the body.
- Many total hip implants operations today rely partially on a porous coating of Hydroxyapatite (HA), a normal component of bone, to help permanently stabilize the stem of the implant in the bone.
- The coating encourages ingrowth from the surrounding tissue that interlocks within the pores much like the pieces of a puzzle lock together.
- Although many current medical procedures **call for inert biocompatible materials**, the increasing understanding of tissue interaction promises many more applications for aggressive bioactive materials.

Commonly Used Biomaterials

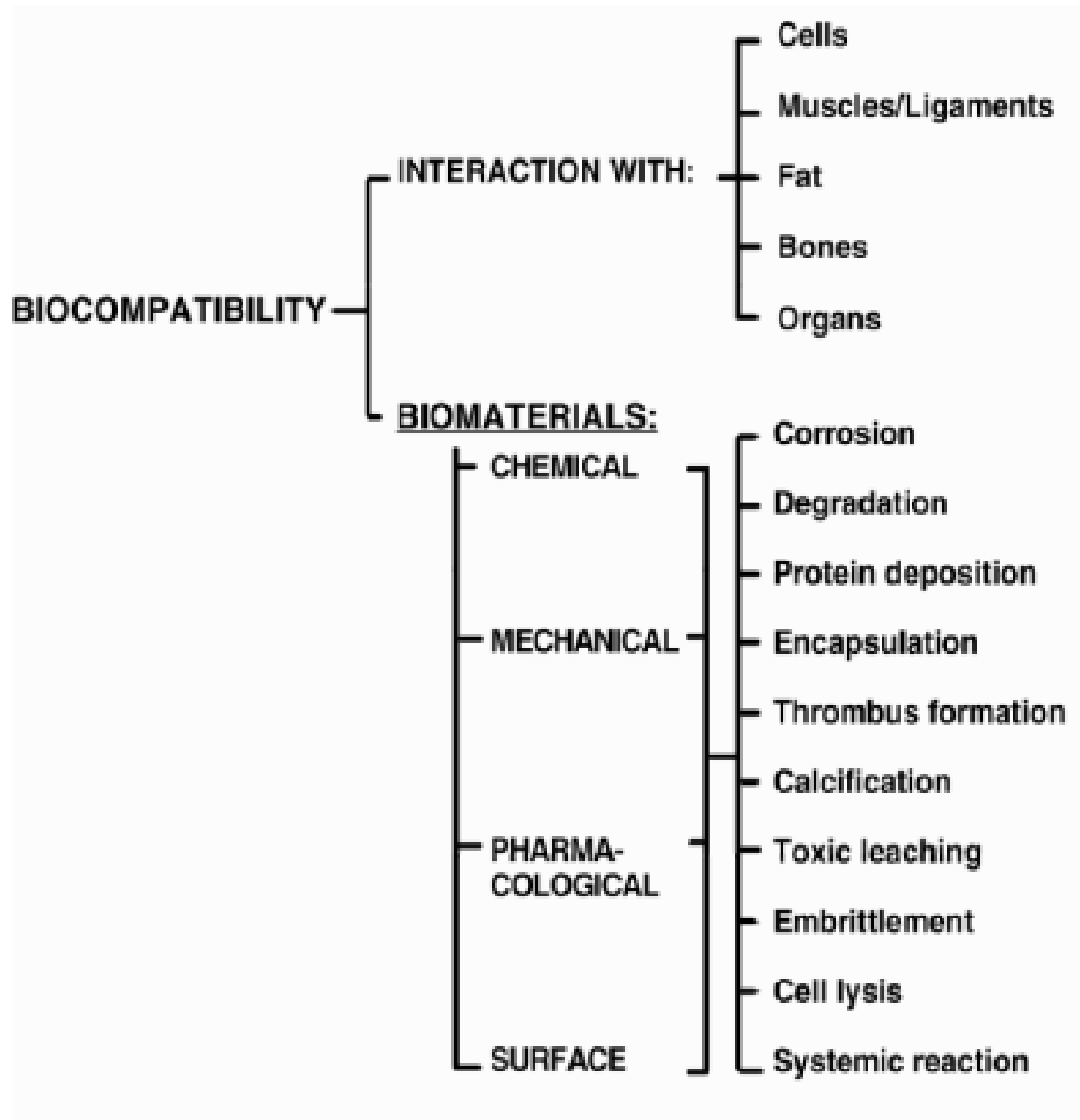
Materials	Applications
Silicone rubber	Catheters, tubing
Dacron (聚酯纤维)	Vascular grafts
Poly(methyl methacrylate)	Intraocular lenses, bone cement
Polyurethanes	Catheters, pacemaker leads
Stainless steel	Orthopedic devices, stents
Collagen (reprocessed) (胶原蛋白)	Cosmetic surgery, wound dressings

Requirements of Biomaterials

A biomaterial must be:

- inert or **specifically interactive**
- biocompatible
- mechanically and chemically stable or
- biodegradable
- processable (for manufacturability)
- nonthrombogenic (if blood-contacting)
- sterilizable

Biocompatibility



Progress in Healthcare

Technology moves on

- from bio-inert & current biocompatible materials with limited useful life
- to 'second generation', structurally and functionally advanced materials
- to body replacement and augmentation devices active in a physiological and pathology-correcting way over an entire life-time
- **Tissue engineering, microsystems and nanotechnology** will address many of these needs

Lecture 2: Various Types of Biomaterials

A: Metals

B: Ceramics

C: Polymers

D: Composites

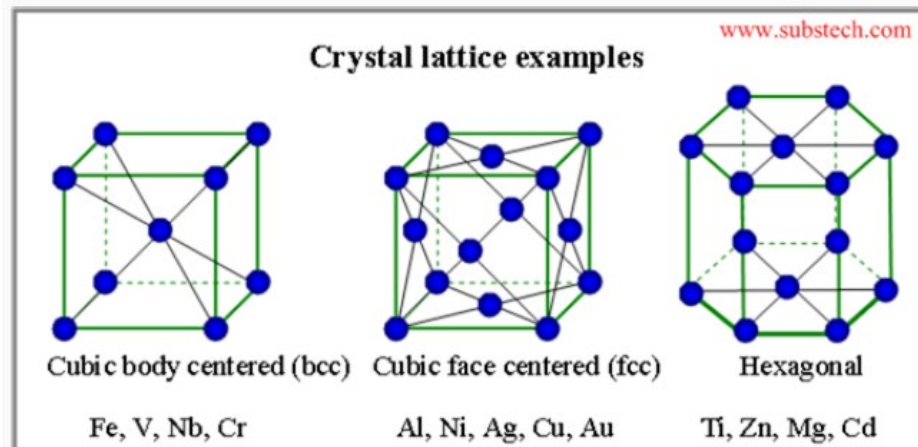
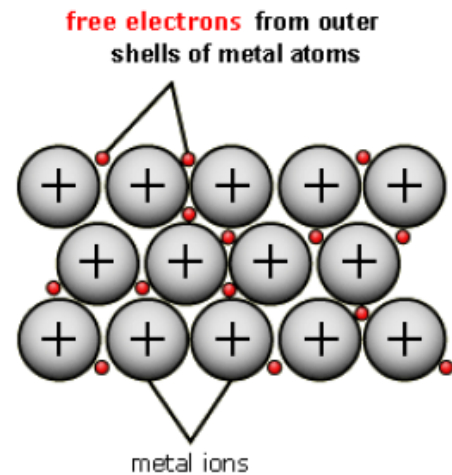
Metals



An overview of metals – bonding characteristics

Metals, Nonmetals, and Metalloids

H																	He				
Li	Be															B	C	N	O	F	Ne
Na	Mg															Al	Si	P	S	Cl	Ar
K	Ca	Sc	Ti	V	Cr	Mn	Fe	Co	Ni	Cu	Zn	Ga	Ge	As	Se	Br	Kr				
Rb	Sr	Y	Zr	Nb	Mo	Tc	Ru	Rh	Pd	Ag	Cd	In	Sn	Sb	Te	I	Xe				
Cs	Ba	La	Hf	Ta	W	Re	Os	Ir	Pt	Au	Hg	Tl	Pb	Bi	Po	At	Rn				
Fr	Ra	Ac	Rf	Ha	Sg	Nc	Hs	Mt										metalloids			
Ce	Pr	Nd	Pm	Sm	Eu	Gd	Tb	Dy	Ho	Er	Tm	Yb	Lu								
Th	Pa	U	Np	Pu	Am	Cm	Bk	Cf	Es	Fm	Md	No	Lr								

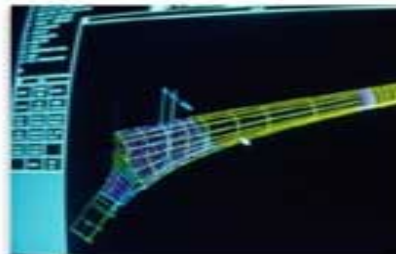


The lattice structure affords high stiffness (modulus) and strength

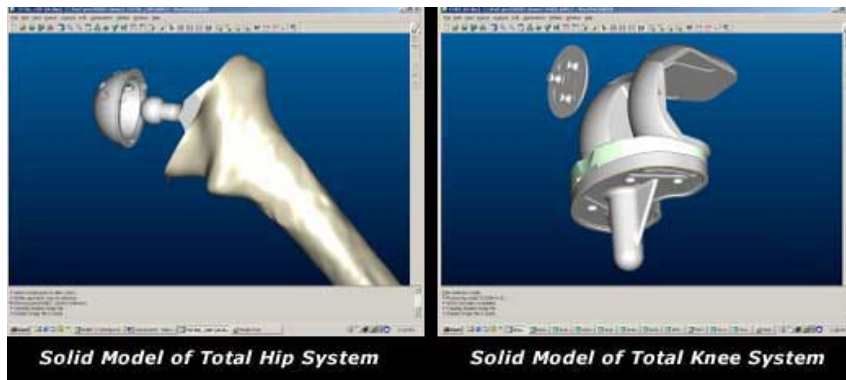
Metals



- closely packed crystal structure; the type of bonding in metals and metal alloys render them valuable as load bearing implants as well as internal fixation devices used for orthopedic applications as well as dental implants;
- when processed suitably they contribute high tensile, fatigue and yield strengths; low reactivity and good ductility to the stems of hip implant devices; and,
- Their properties depend on the processing method and purity of the metal, however, **and the selection of the material must be made appropriate to its intended use.**

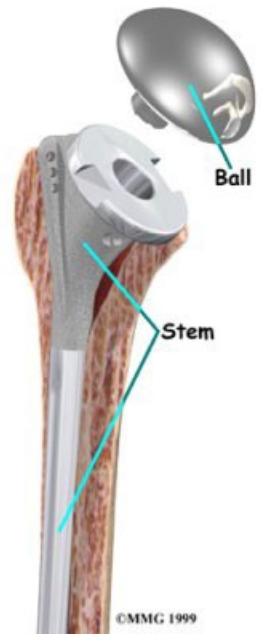


Metals Manufacturing



Metals

- One complication that can occur from the use of metals in orthopedic applications is **the phenomenon of stress shielding**;
- In some situations, such as hip implantation, the high strength of the metal in the implant induces it to assume more than its share of responsibility for the load in that region;
- This decreases the load born by the surrounding tissue and therefore shields it from experiencing stress;
- **Lack of stress causes bone density to decrease as bone tissue resorbs, eventually causing complications in the implant/tissue interface.**



Other Uses of Metals



Medical Tubing



Catheters



Stents

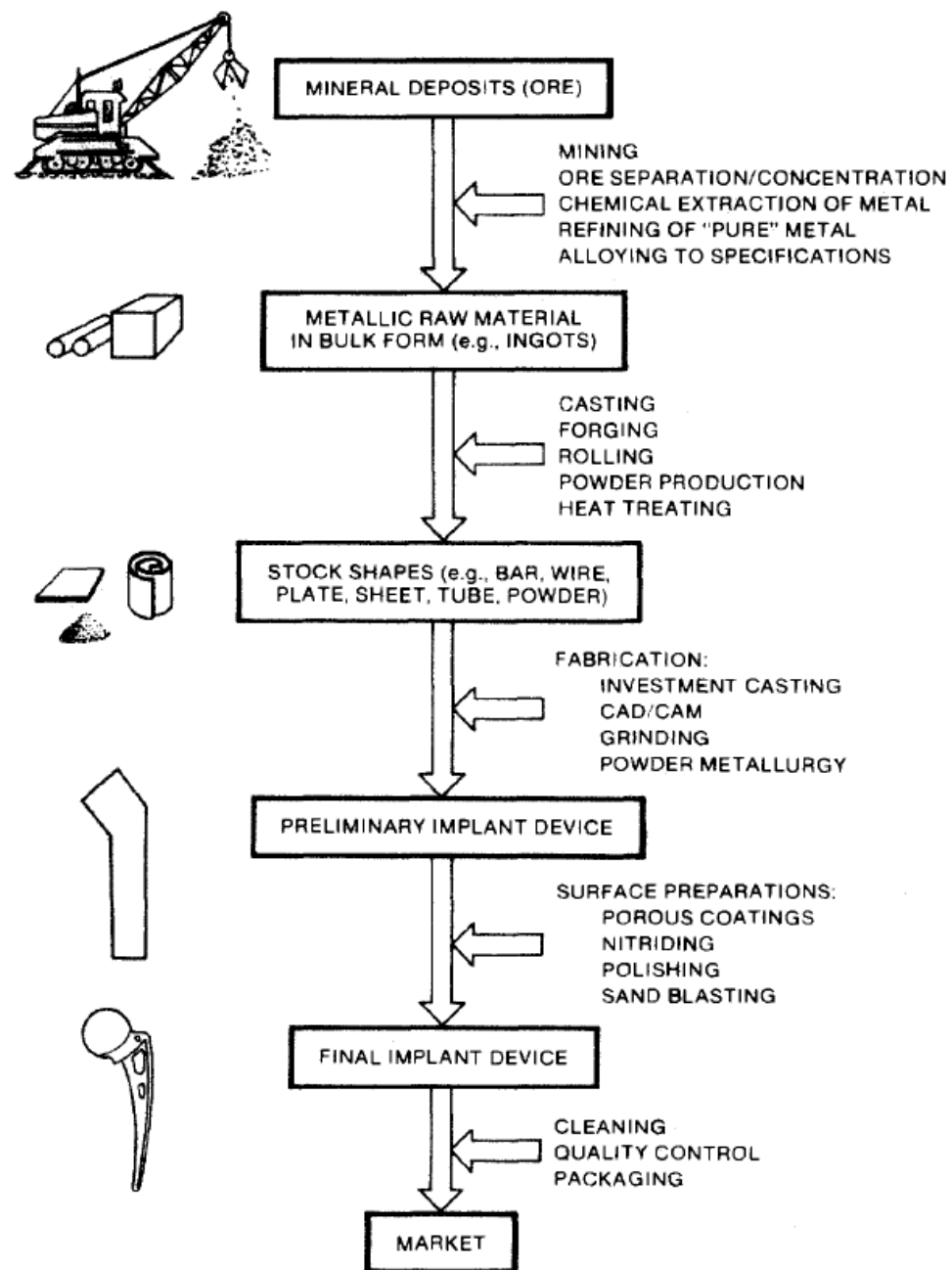
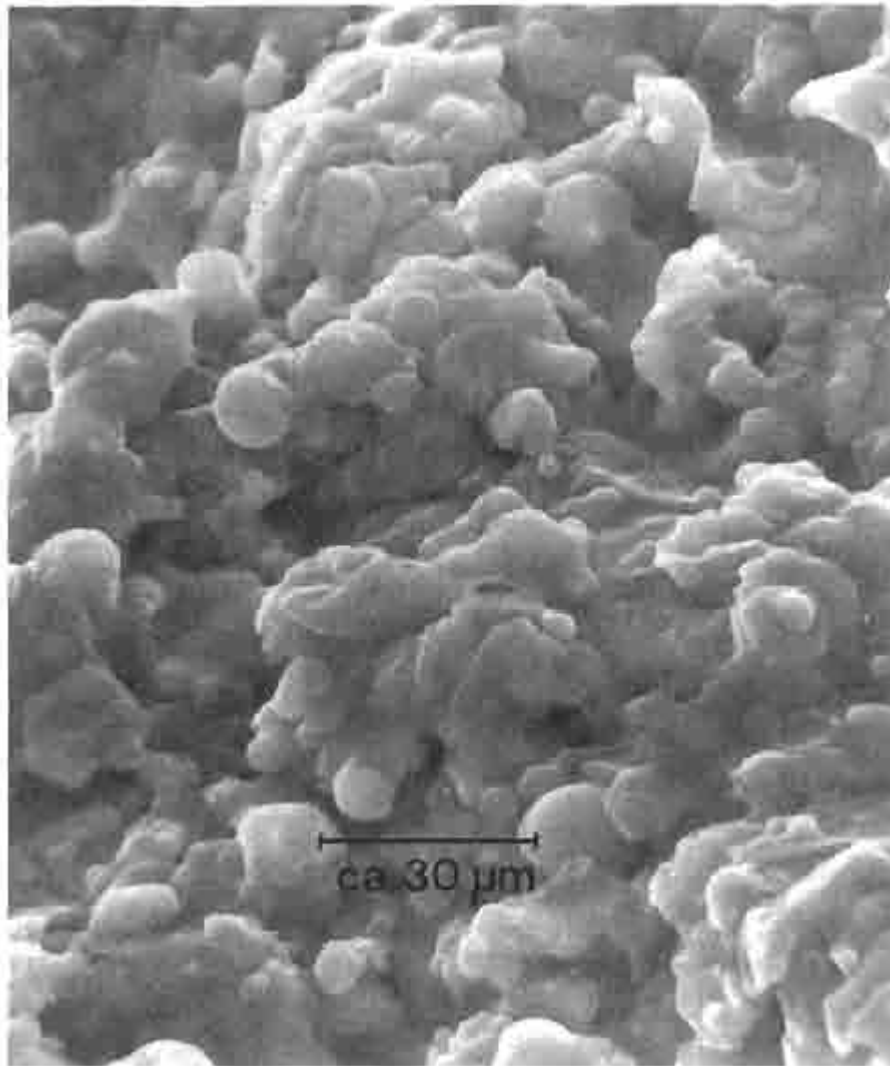


FIG. 1. General processing history of a typical metallic implant device, in this case a hip implant.

TABLE 1 Chemical Compositions of Stainless Steels Used for Implants

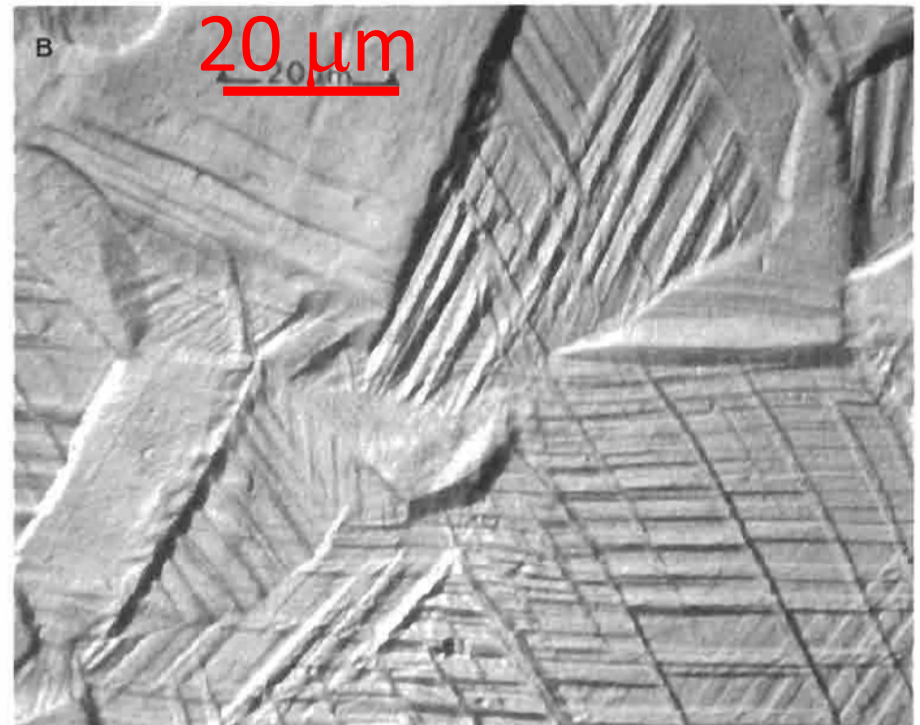
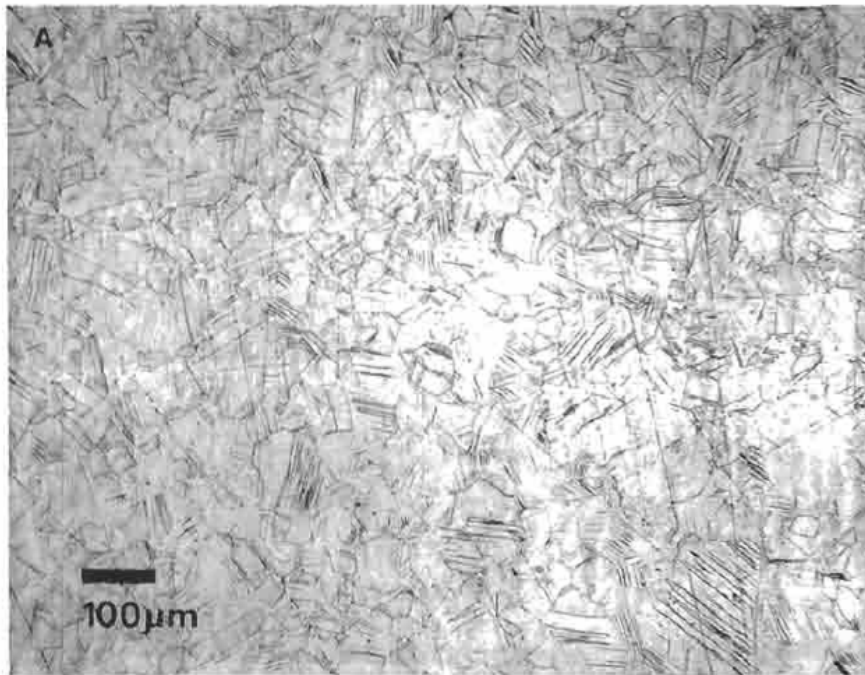
Material	ASTM designation	Common/trade names	Composition (wt. %)	Notes
Stainless steel	F55 (bar, wire)	AISI 316LVM	60–65 Fe	F55, F56 specify 0.03 max for P, S. F138, F139 specify 0.025 max for P and 0.010 max for S. LVM = low vacuum melt.
	F56 (sheet, strip)	316L	17.00–19.00	
	F138 (bar, wire)	316L	12.00–14.00	
	F139 (sheet, strip)	316L	2.00–3.00 max 2.0 Mn max 0.5 Cu max 0.03 C max 0.1 N max 0.025 P max 0.75 Si max 0.01 S	
Stainless steel	F745	Cast stainless steel cast 316L	60–69 Fe 17.00–19.00 11.00–14.00 2.00–3.00 max 0.06 C max 2.0 Mn max 0.045 P max 1.00 Si max 0.030 S	

American Society for Testing and Materials: Standards are documents that serve to formalize procedures, rules, or guidelines for industry to follow. A document that has been developed and established within the consensus principles of **ASTM** International, and which meets the approval requirements of ASTM procedures and regulations, can become one of ASTM International's standards.



Scanning
electron
micrograph of
a titanium
plasma spray
coating on an
oral implant

FIG. 3. Scanning electron micrograph of a titanium plasma spray coating on an oral implant. (Photo courtesy of A. Schroeder, E. Van der Zypen, H. Stich, and F. Sutter, *Int. J. Oral Maxillofacial Surgery* 9:15, 1981.)



A: Typical microstructures of cold-worked 316L stainless steel, ASTM F318 in a transverse section taken through a spinal distraction rod. B: detail of grains in cold-worked 316L stainless steel showing evidence of plastic deformation.

FIG. 4. (A) Typical microstructure of cold-worked 316L stainless steel, ASTM F318, in a transverse section taken through a spinal distraction rod. (B) Detail of grains in cold-worked 316L stainless steel showing evidence of plastic deformation. (Photo courtesy of Zimmer USA, Warsaw, IN.)

TABLE 2 Typical Mechanical Properties of Implant Metals^a

Material	ASTM designation	Condition	Young's modulus (GPa)	Yield strength (MPa)	Tensile strength (MPa)	Fatigue endurance limit (at 10 ⁷ cycles, R = -1) (MPa)
Stainless steel	F745	Annealed	190	221	483	221–280
	F55, F56, F138, F139	Annealed	190	331	586	241–276
		30% Cold worked	190	792	930	310–448
		Cold forged	190	1213	1351	820
Co–Cr alloys	F75	As-cast/annealed	210	448–517	655–889	207–310
		P/M HIP ^b	253	841	1277	725–950
	F799	Hot forged	210	896–1200	1399–1586	600–896
	F90	Annealed	210	448–648	951–1220	Not available
		44% Cold worked	210	1606	1896	586
	F562	Hot forged	232	965–1000	1206	500
		Cold worked, aged	232	1500	1795	689–793
						(axial tension R = 0.05, 30 Hz)
Ti alloys	F67	30% Cold-worked Grade 4	110	485	760	300
	F136	Forged annealed	116	896	965	620
		Forged, heat treated	116	1034	1103	620–689

^aData collected from references noted at the end of this chapter, especially Table 1 in Davidson and Georgette (1986).^bP/M HIP = Powder metallurgy product, hot-isostatically pressed.

TABLE 3 Chemical Compositions of Co-Based Alloys for Implants

Material	ASTM designation	Common/trade names	Composition (wt %)	Notes
Co-Cr-Mo	F75	Vitallium Haynes-Stellite 21 Protasul-2 Micrograin-Zimaloy	58.9–69.5 Co 27.0–30.0 Cr 5.0–7.0 Mo max 1.0 Mn max 1.0 Si max 1.0 Ni max 0.75 Fe max 0.35 C	Vitallium is a trademark of Howmedica, Inc. Haynes-Stellite 21 (HS 21) is a trademark of Cabot Corp. Protasul-2 is a trademark of Sulzer AG, Switzerland. Zimaloy is a trademark of Zimmer USA.
Co-Cr-Mo	F799	Forged Co-Cr-Mo Thermomechanical Co-Cr-Mo FHS	58–59 Co 26.0–30.0 Cr 5.0–7.00 Mo max 1.00 Mn max 1.00 Si max 1.00 Ni max 1.5 Fe max 0.35 C max 0.25 N	FHS means “forged high strength” and is a trademark of Howmedica, Inc.
Co-Cr-W-Ni	F90	Haynes-Stellite 25 Wrought Co-Cr	45.5–56.2 Co 19.0–21.0 Cr 14.0–16.0 W 9.0–11.0 Ni max 3.00 Fe 1.00–2.00 Mn 0.05–0.15 C max 0.04 P max 0.40 Si max 0.03 S	Haynes-Stellite 25 (HS25) is a trademark of Cabot Corp.
Co-Ni-Cr-Mo-Ti	F562	MP 35 N Biophase Protasul-10	29–38.8 Co 33.0–37.0 Ni 19.0–21.0 Cr 9.0–10.5 Mo max 1.0 Ti max 0.15 Si max 0.010 S max 1.0 Fe max 0.15 Mn	MP35 N is a trademark of SPS Technologies, Inc. Biophase is a trademark of Richards Medical Co. Protasul-10 is a trademark of Sulzer AG, Switzerland

TABLE 4 Chemical Compositions of Ti-Based Alloys for Implants

Material	ASTM designation	Common/trade names	Composition (wt %)	Notes
Pure Ti	F67	CP Ti	Balance Ti max 0.10 C max 0.5 Fe max 0.0125–0.015 H max 0.05 N max 0.40 O	CP Ti comes in four grades according to oxygen content—only Grade 4 is listed.
Ti–6Al–4V	F136	Ti–6Al–4V	88.3–90.8 Ti 5.5–6.5 Al 3.5–4.5 V max 0.08 C 0.0125 H max 0.25 Fe max 0.05 N max 0.13 O	

Necessary information for characterization of a metallic biomaterial

Chemical compositions

Mechanical properties

Surface morphologies

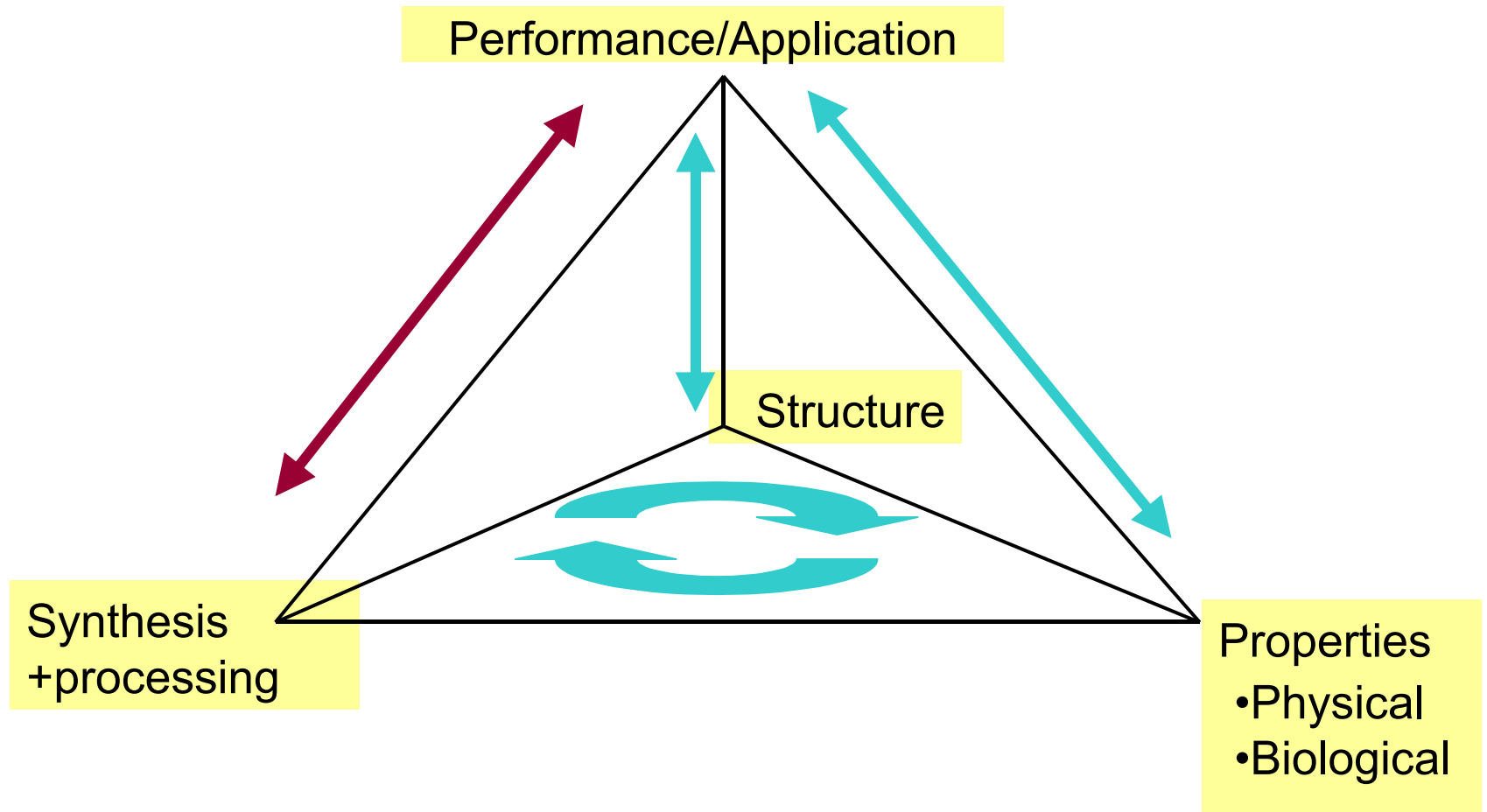
Other characterizations:

Biocompatibility?

Lifetime?

Cost?

Material Science Logic



Ceramics

- Inorganic compounds that contain metallic and non-metallic elements, for which inter-atomic bonding is ionic or covalent, and which are generally formed at high temperatures.
- Derivation: From the Greek word "*keramos*" meaning the art and science of making and using solid articles formed by the action of heat on earthy raw materials.
- Most ceramics occur as minerals:
- (1) The abundance of elements and geochemical characteristics of the earth's crust govern mineral types.
- (2) Composition of *Earth's Crust*. [84% = O + Si + Al]
 - O = 50% Fe = 5% K = 2.5%
 - Si = 26% Ca = 3% Mg = 2%
 - Al = 8% Na = 2.5% H = 1%

Ceramics

- **Advantages:**
 - inert in body (or bioactive in body)
 - high wear resistance (orthopedic & dental applications)
 - high modulus (stiffness) & compressive strength
 - fine esthetic properties for dental applications
- **Disadvantages:**
 - brittle (low fracture resistance, flaw tolerance)
 - low tensile strength (fibers are exception)



Ceramic Applications

- femoral heads and cup inserts for ceramic on polyethylene; or
- ceramic on ceramic hip replacement bearings;
- knee prostheses;
- spinal fusion devices;
- orthopedic instrumentation;
- dental-crowns;
- bridges, implants and caps;
- inner ear implants (cochlear implants);
- drug delivery devices; and,
- cochlear implants.



Ceramics

- Alumina, Zirconium, Calcium phosphate, Silica, pyrolytic carbon, hydroxyapatite are common;
- Porous ceramic materials exhibit much lower strengths but have been found extremely useful as coatings for metallic implants;
- The coating aids in tissue fixation of the implant by providing a porous surface for the surrounding tissue to grow into and mechanically interlock; and,
- Certain ceramics are considered bioactive ceramics if they establish bonds with bone tissue.

TABLE 2 Types of Bioceramic–Tissue Attachment and Their Classification

Type of attachment	Example
1. Dense, nonporous, nearly inert ceramics attach by bone growth into surface irregularities by cementing the device into the tissues or by press-fitting into a defect (termed “morphological fixation”).	Al ₂ O ₃ (single crystal and polycrystalline)
2. For porous inert implants, bone ingrowth occurs that mechanically attaches the bone to the material (termed “biological fixation”).	Al ₂ O ₃ (polycrystalline) Hydroxyapatite-coated porous metals
3. Dense, nonporous surface-reactive ceramics, glasses, and glass-ceramics attach directly by chemical bonding with the bone (termed “bioactive fixation”).	Bioactive glasses Bioactive glass-ceramics Hydroxyapatite
4. Dense, nonporous (or porous) resorbable ceramics are designed to be slowly replaced by bone.	Calcium sulfate (plaster of paris) Tricalcium phosphate Calcium-phosphate salts

TABLE 4 Physical Characteristics of Al_2O_3 Bioceramics

	High alumina ceramics	ISO standard 6474
Alumina content (% by weight)	>99.8	≥ 99.50
Density (g/cm^3)	>3.93	≥ 3.90
Average grain size (μm)	3–6	<7
Ra (μm) ^a	0.02	
Hardness (Vickers hardness number, VHN)	2300	>2000
Compressive strength (MPa)	4500	
Bending strength (MPa) (after testing in Ringer's solution)	550	400
Young's modulus (GPa)	380	
Fracture toughness (K_{IC}) ($\text{MPa}^{1/2}$)	5–6	
Slow crack growth	10–52	

^aSurface roughness value.

TABLE 5 Composition of Bioactive Glasses and Glass-Ceramics (in Weight Percent)

	4S5S Bioglass	4S5S5F Bioglass	4S5S5.4F Bioglass	40S5B5 Bioglass	52S4.6 Bioglass	55S4.3 Bioglass	KGC Ceravital	KGS Ceravital	KGy213 Ceravital	A-W GC	MB GC
SiO ₂	45	45	45	40	52	55	46.2	46	38	34.2	19-52
P ₂ O ₅	6	6	6	6	6	6				16.3	4-24
CaO	24.5	12.25	14.7	24.5	21	19.5	20.2	33	31	44.9	9-3
Ca(PO ₃) ₂							25.5	16	13.5		
CaF ₂		12.25	9.8							0.5	
MgO							2.9			4.6	5-15
MgF ₂											
Na ₂ O	24.5	24.5	24.5	24.5	21	19.5	4.8	5	4		3-5
K ₂ O							0.4				3-5
Al ₂ O ₃									7		12-33
B ₂ O ₃				5							
Ta ₂ O ₅ /TiO ₂									6.5		
Structure	Glass	Glass	Glass	Glass	Glass		Glass- Ceramic	Glass- Ceramic		Glass- Ceramic	Glass- Ceramic
Reference	Hench <i>et al.</i> (1972)	Hench <i>et al.</i> (1972)	Hench <i>et al.</i> (1972)	Hench <i>et al.</i> (1972)	Hench <i>et al.</i> (1972)	Hench <i>et al.</i> (1972)	Gross <i>et al.</i> (1988)	Gross <i>et al.</i> (1988)		Nakamura <i>et al.</i> (1985)	Höhländ and Vogel (1993)

TABLE 6 Present Uses of Bioceramics

Orthopedic load-bearing applications Al_2O_3	Coatings for tissue ingrowth (Cardiovascular, orthopedic, dental and maxillofacial prosthetics) Al_2O_3
Coatings for chemical bonding (Orthopedic, dental and maxillofacial prosthetics) HA Bioactive glasses Bioactive glass-ceramics	Temporary bone space fillers Tricalcium phosphate Calcium and phosphate salts
Dental implants Al_2O_3 HA Bioactive glasses	Periodontal pocket obliteration HA HA-PLA composite Trisodium phosphate Calcium and phosphate salts Bioactive glasses
Alveolar ridge augmentations Al_2O_3 HA HA-autogenous bone composite HA-PLA composite Bioactive glasses	Maxillofacial reconstruction Al_2O_3 HA HA-PLA composite Bioactive glasses
Otolaryngological Al_2O_3 HA Bioactive glasses Bioactive glass-ceramics	Percutaneous access devices Bioactive glasses Bioactive composites
Artificial tendon and ligament PLA-carbon fiber composite	Orthopedic fixation devices PLA-carbon fibers PLA-calcium/phosphorous-base glass fibers

Hydroxyapatite (HA) 羟基磷灰石

HA: D-葡萄糖醛酸及N-乙酰葡萄糖胺组成的双糖单位**玻尿酸**（**Hyaluronan**），又称糖醛酸、**透明质酸**，基本结构是由两个双糖单位D-葡萄糖醛酸及N-乙酰葡萄糖胺组成的大型多糖类

Quiz 1: What is the advantage of using AFM for biological studies

Homework: Why there is a stress shielding? What is its problem?

Homework: what are biomaterials? List 3 kinds of biomaterials and discuss each one's advantages and/or disadvantages

Homework: Find one kind of superhydrophobic and superhydrophilic materials.